NGUYEN QUANG DUY

APPLIED AI ENGINEER | AUTOMATION, DATA & MARTECH

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APPLIED AI ENGINEERING PROJECTS

Personal projects developed with AI coding assistance; methods, boundaries and validation are stated below.

Vietnam Career Agent Kit | AI-assisted Python / MCP / LangChain Core

- Maintain a Python toolkit that separates reviewed source evidence, generated document wording and persistent SQLite application state.
- Retrieve confirmed fact quotes through SQLite FTS5/BM25 and retain source IDs and hashes; this is lexical retrieval rather than an embedding/vector index.
- Expose local MCP tools for career context and observed measurements, keeping retrieval separate from account access and application-execution authority.
- Compose an optional LCEL chain using RunnableLambda stages for retrieval, an explicit host drafting callback and revalidation; copy context before invoking the callback and limit chain concurrency to one.
- Rebuild the current writing brief after generation and compare its fingerprint; reject changed context, unknown fact IDs, duplicate citations and CV bullets that reference more than one source fact.
- Time retrieval, host drafting and revalidation separately; disable ambient LangChain/LangSmith tracing and leave unavailable host token/cost measurements unknown.

Portfolio AI Guide | AI-assisted Node.js / Groq application

- Built a Node.js guide that calls Groq's OpenAI-compatible chat-completions endpoint with openai/gpt-oss-20b; configured an 800-token completion limit and a 15-second abort timeout.
- Requested strict JSON-schema output with answer, citation IDs, suggestions and motion; resolved citation IDs only against the selected source records and returned a fallback for missing citations or invalid answer fields.
- Validated allowed origins, POST/JSON requests, a 400-character question limit and at most four recent turns of 500 characters each before model invocation.
- Returned the leading approved source excerpt when generation failed after retrieval; kept the answer and source link available without fabricating an AI response.
- Defined 24 evaluation cases covering factual grounding, unavailable evidence, false historical context, prompt injection and English/Vietnamese limitations; fixtures specify expected source IDs and response checks.

EXPERIMENTAL LOCAL FINE-TUNING

Qwen3 LoRA | AI-assisted technical experiment

- Ran a one-step CPU Qwen3-0.6B LoRA smoke trial on fictional fixtures using PEFT: rank 8, alpha 16, dropout 0.05 and q_proj/v_proj targets.
- Compared held-out assistant-token loss separately from paired base/adapter generation checks; recorded raw outputs, model/adapter identities and observed token/time measurements.
- Kept experimental wording outside approved CVs when strict JSON/source-link checks failed; no model wording was automatically promoted.

ADDITIONAL DATA PROJECT

Paid professor database project: Python and SQL database-building work processing approximately one billion records in total. Project-based work began in March 2022.

PROFESSIONAL EXPERIENCE

DUYTRUYENTHONG Agency | Founder and Operator | Feb 2024 - Present

- Translate client objectives into websites, content systems and digital workflows; define website functionality, commerce requirements and the scope of digital delivery.
- Develop AI-assisted product prototypes: STOS for sales portfolios and traffic-to-sales workflows, and Cho Kenh as a TikTok marketplace/community concept.
- Build TikTok and Facebook social assets for an F&B client and provide ongoing social-asset management and marketing SaaS services. The client continued the service and purchased additional services.

NINA Web Service | Solution Consultant | Dec 2023 - Dec 2024

- Conducted client discovery to capture business objectives, operational constraints and user needs, using these requirements to define the problems the proposed website should address.
- Translated business requirements into website architecture, user-experience needs and expected functionality, making the intended solution behavior explicit for implementation.
- Prepared developer-facing implementation requirements and website functionality specifications, converting client needs into scoped technical work for handoff to development teams.
- Coordinated clients and developers across planning and delivery, aligning technical feasibility, expected functionality and delivery scope with the requirements gathered from the client.
- Recommended practical digital solutions for online presence, customer engagement and business visibility, keeping recommendations connected to the client requirements.

NINA Web Service | Web Developer | Jun 2023 - Dec 2023

- Developed responsive React.js/JavaScript interfaces and maintained existing web applications, with attention to user-facing functionality, usability and maintainable implementation.
- Built front-end components and supporting application functionality; investigated defects, tested changes and integrated updates into existing applications.
- Made iterative improvements to application functionality and interfaces as part of the development process.
- Prototyped a knowledge-based support chatbot using retrieval-augmented generation and AI-agent concepts for user assistance.

SELECTED NINA DATA-PIPELINE CONTRIBUTION

Databricks | Data-lake transformation and reporting validation

- Implemented a Databricks pipeline that cleans and transforms raw data-lake inputs into structured reporting tables.
- Reconciled source and transformed-table row counts and checked duplicate records and missing values to validate the reporting pipeline outputs.
- Cross-checked sample output records against source data to verify reporting consistency.
- Inspected pipeline job logs and repeated executions, checking that reruns produced consistent reporting results.

PROFESSIONAL EXPERIENCE CONTINUED

iTechWX | Microsoft Technical Support Specialist | Apr 2025 - Feb 2026

- Provided enterprise Microsoft 365 support across OneDrive, SharePoint, Exchange Online and Teams, investigating incidents affecting collaboration services and daily business operations.
- Troubleshot technical issues for enterprise partners involving Microsoft Intune, intranet and server environments.
- Reproduced reported behavior, checked relevant service configurations and tested diagnostic hypotheses to narrow incident causes and document technical evidence.
- Investigated authentication, licensing, synchronization and permissions incidents, linking reported symptoms to the affected Microsoft 365 service and its configuration.
- Analyzed email-delivery incidents with Message Trace and MX/DNS verification; used Microsoft Admin Center and PowerShell to test service-configuration hypotheses and document findings.
- Managed enterprise incidents from intake through customer communication, technical documentation and resolution tracking; coordinated product-level escalations with Microsoft engineering teams.
- Verified service behavior following resolution and reviewed recurring issue patterns; contributed reusable troubleshooting knowledge for consistent handling of future cases.

Amazing Tech | Junior Front-End Developer | Sep 2022 - Jan 2023

- Contributed front-end refactoring and responsive UI implementation alongside senior developers.
- Supported debugging and code-maintainability improvements under senior-developer guidance.

TECHNICAL SKILLS

Programming / web: Python, SQL, JavaScript, Node.js and React.js.

Applied AI: Groq API, LangChain Core/LCEL, MCP, prompt/context design, structured outputs and source-grounded retrieval.

Analysis / automation: Pandas, scikit-learn, n8n and Power Automate; SQL joins/CTEs/window functions and dashboard building with AI assistance.

DIGITAL MARKETING & MARTECH CONTEXT

Hands-on paid advertising since 2023 across Meta Ads, Google Ads, TikTok Ads, Shopee Ads, AppLovin and Pinterest, including mobile-app user acquisition.

Worked with advertising budgets of USD 2,000-10,000 per month per product.

Hands-on use of GA4, Google Tag Manager and AppsFlyer.

EDUCATION AND LANGUAGES

Bachelor of Computer Science | Vietnamese-German University | Completed.

Vietnamese: fluent; English: IELTS 6.5; German: A2; Chinese: HSK 2. HCMC-based Vietnamese citizen; available 1 December 2026.

ADDITIONAL INFORMATION

Claude and ChatGPT: 10,000 hours of hands-on use in total.